

Analyzing and Predicting Student Academic Performance

Motivation

The main motivation behind this project was to apply the concepts learned in DSA 210 to a complete data science workflow on a topic that is both understandable and relevant to daily life. Student academic performance was selected as the project topic because it is influenced by multiple behavioral and educational factors such as attendance, classroom participation, discussion activity, and learning resource usage. This makes it a suitable problem for exploratory data analysis, statistical testing, and machine learning.

Another reason for choosing this topic was that it provides a practical setting in which data science methods can be interpreted clearly. Rather than working on a highly abstract problem, I wanted to focus on a setting where the variables are intuitive and the outcomes are meaningful. The project also gave me the opportunity to practice the full pipeline of a data science study, including data collection, preprocessing, visualization, hypothesis testing, model building, and interpretation of results.

Data Source

The main dataset used in this project is xAPI-Edu-Data, a publicly available Kaggle dataset that contains student demographic, behavioral, and academic performance variables. Kaggle describes this dataset as consisting of 480 student records and 16 features, grouped into demographic, academic background, and behavioral categories. The target of the dataset is student academic performance, usually represented in Low, Medium, and High categories, which makes it suitable for a multiclass classification problem (Aljarah, Kaggle Dataset).

The dataset initially contained 480 observations and 17 variables in my working file, including both categorical and numerical features. During preprocessing, I found 2 duplicate rows, which were removed. After this cleaning step, the working dataset contained 478 observations. The dataset includes variables such as gender, nationality, stage level, topic, semester, relation, raised hands, visited resources, announcement views, discussion activity, parent survey response, parent school satisfaction, absence days, and the final student performance class.

A second dataset, StudentsPerformance.csv, was also added to the repository for possible reference or comparison, but it was not used in the current analysis. The main analysis and all machine learning stages were performed only on xAPI-Edu-Data.csv.

While planning the project, I also reviewed publicly available Kaggle notebook examples related to student academic performance analysis and prediction in order to understand common analysis flows and possible modeling approaches. In particular, I looked at Predicting Student Academic Performance and Student Academic Performance Analysis as reference material during topic exploration and planning, but the final preprocessing, analysis, modeling, and interpretation in this project were carried out independently.

Data Analysis

1. Data Collection and Initial Inspection

The dataset was downloaded from Kaggle and imported into Python for analysis. After loading the data, I examined the shape of the dataset, variable names, data types, summary statistics, and missing values. This helped me understand the structure of the dataset before applying any further steps.

The initial inspection showed that the dataset had no missing values, which simplified the preprocessing stage. However, duplicate records were identified and removed to ensure a cleaner dataset for subsequent analysis.

2. Data Cleaning and Preprocessing

After removing duplicate rows, I prepared the data for both exploratory analysis and machine learning. Since the target variable was represented by the labels L, M, and H, I also created more interpretable labels (Low, Middle, and High) during the exploratory stage.

For the machine learning stage, the target variable was kept as Class, while the remaining columns were used as input features. Because the dataset included many categorical variables, I applied one-hot encoding to transform them into a numerical format suitable for classification models. The target labels were converted into numerical values using Label Encoding. Finally, the dataset was divided into training and test sets using an 80/20 split with stratification in order to preserve class distribution.

3. Exploratory Data Analysis

In the EDA stage, I examined both the target distribution and the relationships between important features and student performance. Several visualizations were used, including count plots, box plots, and a correlation heatmap. These plots focused on variables such as student performance class, gender, absence days, raised hands, visited resources, discussion activity, parent survey response, and parent school satisfaction.

The EDA stage suggested that attendance, classroom engagement, and learning resource usage were strongly related to academic performance. In particular, students with fewer absences, higher classroom participation, and higher resource usage generally appeared in the middle and high performance groups more often.

4. Hypothesis Testing

To move beyond visual impressions, I applied statistical hypothesis testing to examine whether some relationships were statistically significant.

The following tests were used:

- Chi-Square Test of Independence for StudentAbsenceDays and Class
- Chi-Square Test of Independence for ParentAnsweringSurvey and Class
- One-Way ANOVA for raisedhands across performance classes
- One-Way ANOVA for VisITedResources across performance classes

The hypothesis testing results showed statistically significant relationships between absence patterns and student performance, between parent survey response and performance, and

between performance groups and the numerical engagement variables. These tests supported the main patterns observed in the exploratory analysis.

5. Machine Learning Methods

In the machine learning stage, I treated the problem as a multiclass classification task, where the goal was to predict whether a student belonged to the low, middle, or high performance group.

I applied three classification models:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Each model was trained on the training set and evaluated on the test set. The evaluation metrics included accuracy, weighted F1 score, and classification report.

Model	Accuracy	Weighted F1 Score
Logistic Regression	0.7292	0.7277
Decision Tree	0.7083	0.7083
Random Forest	0.8021	0.8021

Among the three models, Random Forest produced the best performance. After identifying the best model, I also examined feature importance using the Random Forest classifier. The most influential features included VisITedResources, raisedhands, StudentAbsenceDays_Above-7, AnnouncementsView, StudentAbsenceDays_Under-7, and Discussion. This result is consistent with the earlier EDA and hypothesis testing stages, where student engagement and attendance already appeared to be key factors associated with academic success.

Findings

This project produced several important findings.

First, student academic performance is strongly associated with classroom engagement and attendance-related variables. Students who visited learning resources more frequently, raised their hands more often, and participated more in discussion activities generally showed better academic performance.

Second, absence days were found to be one of the strongest indicators of performance. Both the visual analysis and the chi-square test suggested that students with more absences were more likely to fall into lower performance groups.

Third, parental involvement also appeared to matter. The relationship between parent survey response and student performance was statistically significant, suggesting that family-related support variables may also have predictive value.

Fourth, machine learning models were able to predict student performance with a reasonably good level of success. Among the tested methods, Random Forest achieved the highest accuracy and weighted F1 score, making it the strongest model in this project.

Finally, the feature importance analysis reinforced the overall interpretation of the project: behavioral engagement variables are among the most useful predictors of academic performance in this dataset.

Limitations and Future Work

Although the project produced meaningful results, it also has several limitations.

One limitation is the size of the dataset. After cleaning, the analysis was based on 478 observations, which is manageable for a course project but still relatively limited for drawing broader conclusions. A larger and more diverse dataset would likely improve model reliability and generalizability.

Another limitation is that the project mainly relied on a single primary dataset. Although the dataset is suitable for classification and educational analysis, future work could benefit from enriching it with additional educational or demographic data sources, which would align more closely with the ideal project design suggested in the course guidelines.

In addition, the machine learning models were applied with limited parameter tuning. More advanced hyperparameter optimization, cross-validation, and additional algorithms such as XGBoost, SVM, or boosting-based methods could improve predictive performance.

A further limitation is that the project focused mainly on predictive performance rather than fairness or bias analysis. Since educational data may contain demographic or structural biases, future work could evaluate whether model performance differs across subgroups and whether fairness issues emerge.

In future work, I would like to include additional educational datasets for enrichment, apply more advanced model tuning techniques, use cross-validation for more robust evaluation, examine fairness and bias across student subgroups, and explore explainability methods such as SHAP to better understand model decisions.

Conclusion

This project followed the main stages of a data science pipeline to analyze and predict student academic performance. Starting from a real-world educational dataset, I carried out data cleaning, exploratory analysis, statistical testing, and machine learning modeling. The results showed that student engagement, attendance, and resource usage are strongly associated with academic success, and that student performance can be predicted with good accuracy using classification models.

Overall, the project helped me apply course concepts in a practical setting and improved my understanding of how data science methods can be used to study meaningful real-world problems.

References

1. Aljarah, I. *Students' Academic Performance Dataset (xAPI-Edu-Data)*. Kaggle. Dataset description notes that the dataset consists of 480 student records and 16 features. Available at: <https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data>
2. *Student Academic Performance competition/dataset context on Kaggle*. Used for understanding the multiclass prediction framing of Low, Medium, and High student performance. Available at: <https://www.kaggle.com/c/student-academic-performance>
3. entropicinertia. *Predicting Student Academic Performance*. Kaggle Notebook. Used as background reference during topic exploration and methodology planning. Available at: <https://www.kaggle.com/code/entropicinertia/predicting-student-academic-performance>
4. bhartiprasad17. *Student Academic Performance Analysis*. Kaggle Notebook. Used as supplementary background reference during the early planning stage. Available at: <https://www.kaggle.com/code/bhartiprasad17/student-academic-performance-analysis>
5. *DSA 210 Introduction to Data Science Project Guidelines*. Used for final report structure, deliverables, and academic integrity requirements.

AI Assistance Disclosure

AI assistance was used after the initial topic-selection stage of this project to support planning, organization, writing, code structuring, debugging, and documentation. More specifically, after the project topic had been selected, I used an LLM-based assistant to help me understand the milestone requirements, organize the GitHub repository, structure the notebooks, improve code clarity, refine parts of the exploratory data analysis and machine learning workflow, and draft the final report.

The assistant was used as a support tool rather than as a replacement for the project work. I personally selected the project topic, reviewed the dataset, executed the notebook cells, checked the outputs, interpreted the results, chose what to include in the final submission, and managed the GitHub milestones and repository updates.

Examples of prompts used during the project include:

- Explain what I need to do for the data collection, EDA, and hypothesis testing milestone.
- Help me structure a notebook for applying machine learning methods to this dataset.
- Compare Logistic Regression, Decision Tree, and Random Forest for this project.
- Help me prepare a final report draft based on my proposal, EDA, hypothesis tests, and ML outputs.
- Help me organize my GitHub repository and milestone tags.

AI-generated outputs included suggested notebook section structures, example Python code blocks for preprocessing, EDA, hypothesis testing, and machine learning, wording suggestions for the proposal, README, and final report, and guidance for GitHub milestone tagging and documentation.

All final execution, validation, interpretation, and submission decisions were made by me.